

Polymarket Weather Rounding & Precision Notes

Concise implementation guidance for PolyTempo

Executive answer

For the Polymarket temperature contracts reviewed, the binding settlement value is the whole-degree Celsius value shown by the linked Weather Underground daily-history station page. The contract rules say the source measures temperatures to whole degrees Celsius. I did not find binding Polymarket rule text saying Polymarket independently downloads hidden decimal values from Wunderground and then rounds them itself.

Operationally: keep forecast values as decimals, but treat the final Weather Underground / Polymarket settlement value as an integer label. For modeling, use a clearly documented nearest-integer assumption; for settlement/backtesting, use the final integer actually shown by the source.

1. Is everything integer-only?

Layer	Precision exposed	How PolyTempo should treat it
Open-Meteo forecast	Decimal / float values, e.g. daily temperature max can be 28.9°C.	Store raw forecasts as floats. Do not round before modeling.
Weather Underground daily history	Daily high/low are displayed as whole numbers on the history page; some other fields, like averages, may show decimals.	Treat displayed daily high/low integer as the settlement-relevant observation.
Polymarket contract	Whole-degree Celsius buckets.	Final result is discrete: one integer bucket wins.

2. Is rounding floor or nearest?

For Open-Meteo, there is no need to round: the API returns decimals, so keep them as decimals.

For Weather Underground settlement, the important point is that the daily high/low value exposed for the contract is already a whole-number value. I did not find primary documentation proving whether Wunderground internally floors, rounds to nearest, or uses another display/aggregation process for the daily high/low integer.

For PolyTempo modeling, use this explicit working assumption:

nearest integer, half-up: 16.4 -> 16, 16.5 -> 17, 16.6 -> 17

But mark this as a modeling assumption. Do not pretend it is a confirmed Wunderground internal rule.

3. Where is rounding done for the Polymarket contract?

The most defensible interpretation is: Weather Underground publishes the settlement-relevant daily high/low as a whole-degree value, and Polymarket resolves according to that source value. The contract rules reviewed state that the resolution source measures temperatures to whole degrees Celsius.

So for implementation:

forecast decimal -> PolyTempo provisional bucket mapping -> wait for wunderground integer -> settle/backtest against that integer

If the true sensor maximum were secretly 16.6°C but Wunderground only displays an integer, the contract effectively follows the displayed integer. If Wunderground displays 17°C, settle/backtest as 17°C. If it displays 16°C, settle/backtest as 16°C. We should not reconstruct a hidden decimal unless the contract explicitly provides it.

4. What does `daily=temperature_2m_max` mean?

In Open-Meteo, `temperature_2m` means air temperature at **2 meters above ground**. It does not mean 2 minutes.

`daily=temperature_2m_max` means the daily maximum of the 2-meter air temperature for the requested location/date. It is a forecast/model variable, usually returned as a decimal Celsius value if `temperature_unit=celsius`.

Important nuance: Open-Meteo is coordinate/model-grid based. Requesting the coordinates of EGLC gives a forecast for that location/grid point, not the literal future reading of the Wunderground airport station sensor.

5. Recommended PolyTempo data policy

Field	Type	Meaning
<code>forecast_raw_c</code>	float	Open-Meteo or future source decimal forecast value.
<code>forecast_bucket_model</code>	int / label	PolyTempo provisional bucket using documented nearest-integer assumption.
<code>settlement_c</code>	int	Final Wunderground / Polymarket whole-degree settlement value.
<code>settlement_raw_c</code>	float null	Only fill if the actual settlement source exposes a raw decimal. Otherwise keep null.

6. Concrete implementation answer

Question: if a forecast says 16.6°C, should PolyTempo map it to 16°C or 17°C?

For modeling: map 16.6°C to 17°C under nearest-integer bucket logic. For final settlement: do not compute this yourself; use the integer shown by Weather Underground / Polymarket settlement source.

This separation prevents a common bug: rounding your forecast value and then assuming that is the same as the eventual settlement value. Forecast bucketing is a model assumption; settlement is a source value.

Source notes

This note is based on reviewed Polymarket temperature-contract rules, Weather Underground daily-history behavior, and Open-Meteo documentation/API examples. The key uncertainty is Wunderground internal quantization of daily high/low: the public daily high/low appears integer-valued, but a modern primary document specifying floor vs nearest was not found.